A negative-parameter counterexample to the unrestricted q-gamma extension

MathDB 335285

Principia Math in Progress

12 September 2026

Task 1: resolve the stated extension. Conjecture 1 of Kalmykov–Karp [1, p. 344] proposes extending all conclusions of their Theorem 3 to every $\alpha > 0$, retaining the other hypotheses. For a nontrivial nonnegative log-concave sequence (g_n) with no internal zeros, that theorem concerns

$$g(\mu; x) = \sum_{n\geq 0} \frac{g_n x^n}{\Gamma_q(\mu + n)}, \qquad \Delta_g = g(\mu+\alpha; x)g(\mu+\beta; x) - g(\mu; x)g(\mu+\alpha+\beta; x).$$

The retained parameter condition is $\beta > 0$ and $\mu \geq \max(-\beta, -1)$ [1, p. 342]. The proposed positivity is false on this full domain. The counterexample below does not decide the separately restricted problem with $\mu \geq 0$.

Theorem 1. *Set*

$$q = \frac{256}{625}, \qquad \mu = -\frac{3}{4}, \qquad \alpha = \frac{1}{4}, \qquad \beta = 2, \qquad g_0 = 1, \quad g_n = 0 \quad (n \geq 1).$$

This sequence satisfies the hypotheses of Theorem 3, its series has radius of convergence infinity, and for every $x > 0$,

$$\boxed{\Delta_g = -\frac{95}{576\,\Gamma_q(1/4)\Gamma_q(1/2)} < 0.}$$

In particular, both the value-positivity and coefficient-positivity conclusions of the proposed unrestricted extension fail.

Proof. For $0 < q < 1$, write $[z]_q = (1-q^z)/(1-q)$. The defining product

$$\Gamma_q(z) = (1-q)^{1-z}\frac{(q;q)_\infty}{(q^z;q)_\infty}, \qquad (a;q)_\infty = \prod_{j\geq 0}(1-aq^j),$$

gives $\Gamma_q(z+1) = [z]_q\Gamma_q(z)$ wherever both sides are finite. For positive real z its product is positive and finite: every factor is positive, and the geometric tails have finite sums. Put $A = 1/\Gamma_q(1/4) > 0$ and $B = 1/\Gamma_q(1/2) > 0$. The four arguments in Δ_g are $-3/4, -1/2, 5/4, 3/2$, so none is a pole. The recurrence gives

$$g(-3/4; x) = [-3/4]_q A, \quad g(-1/2; x) = [-1/2]_q B, \quad g(5/4; x) = \frac{A}{[1/4]_q}, \quad g(3/2; x) = \frac{B}{[1/2]_q}.$$

Consequently,

$$\frac{\Delta_g}{AB} = \frac{[-1/2]_q}{[1/4]_q} - \frac{[-3/4]_q}{[1/2]_q} = \frac{1-25/16}{1-4/5} - \frac{1-125/64}{1-16/25} = -\frac{45}{16} + \frac{1525}{576} = -\frac{95}{576}.$$

Here $q^{1/4} = 4/5$ was used. The sequence has support $\{0\}$, is nontrivial, and satisfies $g_n^2 \geq g_{n-1}g_{n+1}$ for every $n \geq 1$; its zeros are all terminal. Moreover, $\mu = -3/4 > -1 = \max(-2, -1)$ and $\alpha > 0$. Thus every retained hypothesis holds, with a strictly negative conclusion. □

The obstruction persists for positive infinite sequences. Take instead $g_n = 1$ for all $n \geq 0$. This is a strictly positive log-concave sequence. For each of the four parameter values above, the product formula gives

$$\lim_{n\to\infty} \left| \frac{1/\Gamma_q(\nu + n + 1)}{1/\Gamma_q(\nu + n)} \right| = 1 - q,$$

so the radius of convergence is $1/(1 - q) = 625/369$. The constant coefficient of Δ_g is the same negative number as in the theorem. Analyticity at $x = 0$ therefore gives $\Delta_g < 0$ for all sufficiently small positive x. The refutation thus also applies with no zero coefficients and with x strictly inside the convergence interval.

An additional finite check. For $g_0 = g_1 = 1$ and $g_n = 0$ for $n \geq 2$, the same calculation gives

$$\frac{\Delta_g}{AB} = -\frac{95}{576} + \frac{3051749269}{3176712000}x + \frac{86127716}{67284525}x^2.$$

At $x = 1/100$ this equals $-15036888769351/96889716000000 < 0$. The accompanying exact-arithmetic verifier checks this polynomial using the four shifted q-gamma recurrences.

Scope and status. The parameter restrictions above were checked in both the current arXiv source and the full published paper, rather than inferred from MathDB's abbreviated theorem references. The failure occurs inside the retained negative-μ range and does not rely on a pole or on setting $x = 0$. It identifies a necessary repair to the domain of the extension. A current literature and public-claim audit found no matching full solution claim; its sources and limitations are recorded in the accompanying audit. This manuscript has not received external peer review.

References

[1] S. I. Kalmykov and D. B. Karp, *Inequalities for series in q-shifted factorials and q-gamma functions*, J. Math. Anal. Appl. **460** (2018), 332–351. doi:10.1016/j.jmaa.2017.11.041. Preprint: arXiv:1702.03788.

Task 2: resolve the nonnegative-parameter version. Determine whether the full value and coefficient conclusions hold for every $\alpha, \beta > 0$ when $\mu \geq 0$ and (g_n) is doubly positive. Begin by checking later claims for this restricted version.

Task 3: determine an admissible negative-parameter range. Find sharp additional restrictions on μ, α, β below $\mu = 0$ that can hold uniformly over all doubly positive coefficient sequences. Assess whether adding $\mu + \min(\alpha, \beta) \geq 0$ is sufficient before attempting a general theorem.

Task 4: classify the two-term case. For $g_0, g_1 > 0$ and $g_n = 0$ for $n \geq 2$, classify when the quadratic $\Delta_g(x)$ is nonnegative throughout $x > 0$. Compare the resulting domain with the conditions needed for nonnegative coefficients.

Task 5: quantify persistence under positive tails. For the counterexample parameters, find the maximal interval adjacent to zero on which $\Delta_g(x) < 0$ for $g_n = 1$. Develop explicit bounds that remain valid for a specified class of positive log-concave perturbations of this sequence.